

$$f(x) = \begin{cases} x/2, & x \in [0, 2] \\ 1, & x \in [2, 4] \end{cases}$$

$$\textcircled{1} L = 4 - 0 = 4$$

$$1, \cos \frac{n\pi x}{2}, \sin \frac{n\pi x}{2}$$

$$f(x) \sim \frac{a_0}{2} + \sum_{n=1}^{\infty} \left(a_n \cos \frac{n\pi x}{2} + b_n \sin \frac{n\pi x}{2} \right)$$

$$a_n = \frac{2}{L} \int_0^L f(x) \cos \frac{n\pi x}{L} dx, \quad b_n = \frac{2}{L} \int_0^L f(x) \sin \frac{n\pi x}{L} dx$$

$$a_n = \frac{1}{2} \int_0^2 f(x) \cos \frac{n\pi x}{2} dx + \frac{1}{2} \int_2^4 f(x) \cos \frac{n\pi x}{2} dx$$

$$a_0 = \frac{1}{2} \int_0^4 f(x) dx = \frac{1}{2} \left(\int_0^2 \frac{x}{2} dx + \int_2^4 1 dx \right) =$$

$$= \frac{1}{2} \left(\frac{1}{2} \cdot \frac{x^2}{2} \Big|_0^2 + \frac{1}{2} \cdot 2 \right) = \frac{1}{4} \cdot 2 + 1 = \frac{3}{2}$$

$$a_0/2 = 3/4 \quad (*) b_n = \frac{1}{4} \int_0^2 x \sin \frac{n\pi x}{2} dx + \frac{1}{2} \int_2^4 \sin \frac{n\pi x}{2} dx$$

$$a_n = \frac{1}{2} \left(\int_0^2 \frac{x}{2} \cos \frac{n\pi x}{2} dx + \int_2^4 \cos \frac{n\pi x}{2} dx \right)$$

$$2d = n\pi/2, \quad a_n = \frac{1}{4} \int_0^2 x \cos \frac{n\pi x}{2} dx + \frac{1}{2} \int_2^4 \cos \frac{n\pi x}{2} dx$$

$$1) u = x, \quad dv = \cos \frac{n\pi x}{2} dx \Rightarrow \int x \cos \frac{n\pi x}{2} dx = \frac{x \sin \frac{n\pi x}{2}}{\frac{n\pi}{2}} - \int \frac{\sin \frac{n\pi x}{2}}{\frac{n\pi}{2}} dx$$

$$= \left(\frac{x \sin \frac{n\pi x}{2}}{\frac{n\pi}{2}} + \frac{\cos \frac{n\pi x}{2}}{\frac{n^2 \pi^2}{4}} \right) \Big|_0^2, \quad 2d = n\pi$$

$$\textcircled{2} \frac{1 - (-1)^{n-1}}{n^2} \quad u \frac{1}{4} \int_0^2 1 - 1 = \frac{(-1)^n - 1}{4n^2} = \frac{(-1)^n - 1}{n^2 \pi^2}$$

$$2) \frac{1}{2} \int_2^4 \cos \frac{n\pi x}{2} dx = \frac{1}{2} \left(\frac{\sin \frac{n\pi x}{2}}{\frac{n\pi}{2}} \right) \Big|_2^4 = 0 \Rightarrow a_n$$

$$* 3) u = x, \quad dv = \sin \frac{n\pi x}{2} dx, \quad \int x \sin \frac{n\pi x}{2} dx = - \frac{x \cos \frac{n\pi x}{2}}{\frac{n\pi}{2}} + \int \frac{\cos \frac{n\pi x}{2}}{\frac{n\pi}{2}} dx$$

$$du = dx, \quad v = - \frac{\cos \frac{n\pi x}{2}}{\frac{n\pi}{2}} \quad \int \frac{\cos \frac{n\pi x}{2}}{\frac{n\pi}{2}} dx =$$

$$= \frac{x \cos \frac{n\pi x}{2}}{\frac{n\pi}{2}} + \frac{\sin \frac{n\pi x}{2}}{\frac{n^2 \pi^2}{4}}$$

$$\frac{1}{4} \left(\frac{1 - (-1)^n}{n^2} \right) = \frac{1 - (-1)^n}{4n^2} = \frac{1 - (-1)^n}{n^2 \pi^2}$$

$$\textcircled{4} \frac{1}{2} \int_0^4 \sin 2x dx = -\frac{\cos 2x}{2} \Big|_0^4 = \frac{\cos 2 - \cos 4}{2} =$$

$$\frac{(-1)^n - 1}{2\alpha} = \frac{1 - (-1)^n}{n\pi}$$

$$a_n = \frac{(-1)^n - 1}{n^2 \pi^2} \quad b_n = + \frac{1 - (-1)^n}{n\pi} - \frac{(-1)^n}{n\pi} = -\frac{1}{n\pi}$$

$$f(x) \sim \frac{3}{4} + \sum_{n=1}^{\infty} \left(\frac{(-1)^n - 1}{n^2 \pi^2} \cos \frac{n\pi x}{2} - \frac{1}{n\pi} \frac{n\pi x}{2} \right)$$

но требуется ряд сход, т.к. ф. непрерыв

$$\lim_{x \rightarrow 2-} f = \lim_{x \rightarrow 2+} f = 1, \quad f(0+) = 0, \quad f(4-) = 1;$$

и макс. период 4 есть шаг и $T(x) = \frac{0+1}{2} = \frac{1}{2}$

$$T(x) = \begin{cases} \frac{x - 4k}{2}, & x \in [4k, 4k+2] \\ 1 & x \in [4k+2, 4k+4] \\ \frac{1}{2} & x = 4k \end{cases}$$

$$\textcircled{2} \quad F(x) = \begin{cases} 1, & x \in [-4, -2] \\ \frac{|x|}{2}, & x \in [-2, 2] \\ 1, & x \in [2, 4] \end{cases}, \quad L=8$$

нужно при sin разл. с, тогда

$$F(x) \sim \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos \frac{n\pi x}{4}$$

$$a_0 = \frac{1}{2} \int_0^4 f(x) dx = \frac{1}{2} \left(\int_0^2 \frac{x}{2} dx + \int_2^4 1 dx \right) = \frac{3}{2}$$

$$a_n = \frac{1}{2} \int_0^4 f(x) \cos \frac{n\pi x}{4} dx, \quad \alpha = n\pi/4$$

$$a_n = \frac{1}{4} \int_0^2 x \cos \alpha x dx + \frac{1}{2} \int_2^4 \cos \alpha x dx$$

$$1) u=x \quad dv=\cos x dx$$

$$du=dx, \quad v=\sin x$$

$$\int_0^{\pi/2} x \cos x dx = \frac{x^2 \sin x}{4} + \frac{\cos x - 1}{4x^2} = \frac{\sin \pi/2}{2\pi} = \frac{\cos \pi/2 - 1}{2\pi}$$

$$2) \frac{1}{2} \int_0^{\pi/2} \cos x dx = \frac{\sin \pi/2 - \sin 0}{2\pi} = -\frac{\sin(\pi/2)}{2\pi}$$

$$a_n = \frac{\cos(n\pi/4) - 1}{4x^2} = 4 \frac{\cos(n\pi/4) - 1}{n^2\pi^2}$$

$$f(x) \sim \frac{3}{4} + \sum_{n=1}^{\infty} 4 \frac{\cos(n\pi/4) - 1}{n^2\pi^2} \cos \frac{n\pi x}{4}$$

$$S(x) = f(x) = \begin{cases} 1, & x \in [-4 + 8k, -2 + 8k] \\ \frac{|x-8k|}{2}, & x \in [-2 + 8k, 2 + 8k] \\ 1, & x \in [2 + 8k, 4 + 8k] \end{cases}$$

$$3) f(x) = \begin{cases} 1, & x \in [-4, -2] \\ \frac{x}{2}, & x \in [-2, 2] \\ 1, & x \in [2, 4] \end{cases} \quad L=8$$

$$a_n = 0, \quad f(x) \sim \sum_{n=1}^{\infty} b_n \sin \frac{n\pi x}{4}$$

$$b_n = \frac{1}{2} \int_{-4}^4 f(x) \sin \frac{n\pi x}{4} dx, \quad x = \frac{n\pi}{4}$$

$$b_n = \frac{1}{4} \int_{-4}^{-2} x \sin \frac{n\pi x}{4} dx + \frac{1}{2} \int_{-2}^2 \sin \frac{n\pi x}{4} dx$$

$$1) \frac{1}{4} \int_{-4}^{-2} x \sin \frac{n\pi x}{4} dx = \frac{1}{4} \left(-\frac{x \cos \frac{n\pi x}{4}}{2} + \frac{\sin \frac{n\pi x}{4}}{2} \right) =$$

$$-\frac{\cos \pi n/2}{2x} + \frac{\sin \pi n/2}{4x^2}$$

$$2) \frac{1}{2} \int_{-2}^2 \sin \frac{n\pi x}{4} dx = -\frac{\cos \pi n/2 - \cos \pi n/2}{2x} = -\frac{\cos \pi n/2 - (-1)^n}{2x}$$

$$b_n = -\frac{\cos \pi n/2}{2x} + \frac{\sin \pi n/2}{4x^2} + \frac{\cos \pi n/2 - (-1)^n}{2x} =$$

$$= \frac{4 \sin \pi n/2}{4 \pi^2 x^2} - \frac{2(-1)^n}{n\pi}$$

$$F(x) \sim \sum_{n=1}^{\infty} \left(\frac{4 \sin(n\pi/2)}{n^2 \pi^2} - \frac{2(-1)^n}{n\pi} \right) \sin \frac{n\pi x}{4}$$

$$T(x+8) = T(x)$$

$$T(x) = \begin{cases} -1, & x \in [-4, -2] \\ \frac{x}{2}, & x \in [-2, 2] \\ 1, & x \in [2, 4] \end{cases}$$

check at $x = 4 + 8k$ and $x = -4 + 8k$

$$T(4-0) = 1 \quad T(-4+0) = -1, \text{ no jump } T = \frac{1+(-1)}{2} = 0$$

$$T(x) = \begin{cases} -1, & x \in (-4 + 8k, -2 + 8k) \\ \frac{x - 8k}{2}, & x \in [-2 + 8k, 2 + 8k] \\ 1, & x \in [2 + 8k, 4 + 8k) \\ 0, & x = 4 + 8k \end{cases}$$

1



$y = -1 \{ -4 < t < -2 \}$

×

2



$y = \frac{t}{2} \{ -2 < = t < = 2 \}$

×

3



$y = 1 \{ 2 < t < 4 \}$

×

4



$y = 0 \{ \text{abs}(t) = 4 \}$

×

5



$k = \text{floor}\left(\frac{(x+4)}{8}\right)$

×

6



$t = x - 8k$

×



1


 $y = 1 \{ -4 < = t < = -2 \}$

2


 $y = \frac{\text{abs}(t)}{2} \{ -2 < = t < = 2 \}$

3


 $y = 1 \{ 2 < = t < 4 \}$

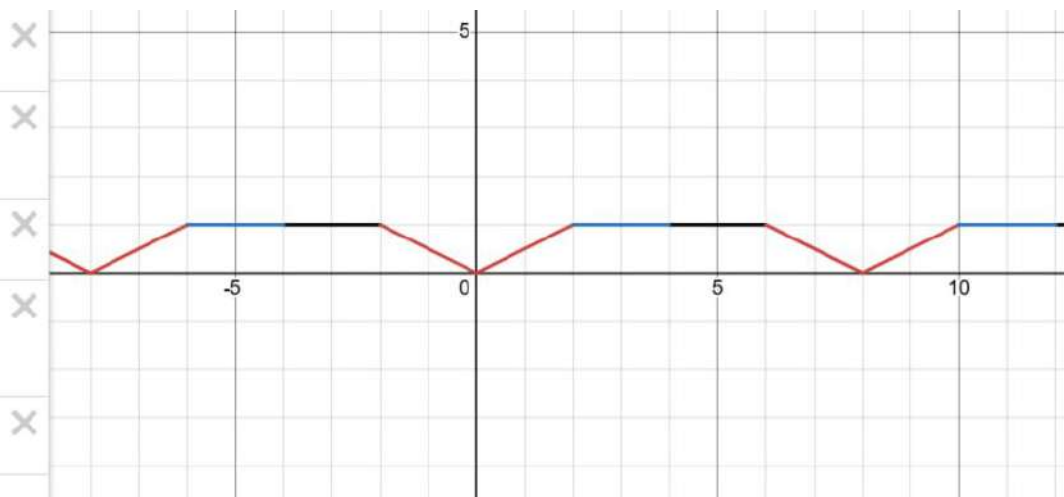
4


 $k = \text{floor}\left(\frac{(x + 4)}{8}\right)$

5


 $t = x - 8k$

6



1



2



3



4



$$y = \frac{(x - 4k)}{2} \{ 0 + 4k < x < 2 + 4k \}$$

$$y = 1 \{ 2 + 4k < = x < 4 + 4k \}$$

$$y = 1/2 \{ x = 4k \}$$

$$k = \text{floor} (x/4)$$

×

×

×

×

